SILVANUS ALVAN

AI RESEARCH FELLOW & MACHINE LEARNING ENGINEER

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SUMMARY

Seventh-semester Informatics student at Universitas Pelita Harapan with a GPA of 3.71, specializing in Artificial Intelligence, Computer Vision, and Full-Stack Software Development. Proven experience in designing deep learning architectures (CNN-GRU, ENet), building cross-platform enterprise apps (Flutter/Firebase), and publishing academic research in NLP and machine learning. Detail-oriented, highly analytical, and eager to contribute to innovative remote engineering teams.

WORK EXPERIENCE

Office Admin, PT. KARYA SATRIA **June 2022 - June 2023**

- Preparing and managing proposals and Work Orders (SPK).
- Handling tax-related matters for company transactions and projects.
- Managing project budgeting to ensure cost efficiency.
- Coordinating procurement of goods and services from vendors and overseeing client orders.
- Experienced in managing projects across Tangerang, Banten, West Jakarta, and South Jakarta as admin project.
- Responsible as supervisor for 3 months.

Part Time Office Admin, PT. KARYA SATRIA **August 2023 - August 2024**

- Preparing and managing proposals and Work Orders (SPK).
- Handling tax-related matters for company transactions and projects.
- Coordinating procurement of goods and services from vendors and overseeing client orders.
- Experienced in managing projects across Tangerang, Banten, West Jakarta, and South Jakarta as admin project.

EDUCATION

August 2023 - Ongoing

Bachelor in Computer Science

Universitas Pelita Harapan

- Specialization in Data Science Curriculum

ADDITIONAL INFORMATION

- **Technical Skills:** C++, CUDA, Python, Dart, Java, JavaScript, Computer Vision, Natural Language Processing, HTML, CSS, PyTorch, Flutter, Full-stack Development, Automa, n8n, LLM, N8N
- **Languages:** English, Mandarin HSK 3
- **Certifications:** EDX Introduction to Game Design: Game Systems and Balance, Process and
- Creation, A Complete Guide to Game Design. CS50's Introduction to Cybersecurity

PROJECTS

ENet-based Lane Segmentation (Enet + SAD + Dual Attention + Temporal Lane Memorization).

- Conducted a systematic ablation study on ENet architectures, achieving a peak F1-score of 56.13% and reducing the False Negative Rate to 20.10%.
- Optimized the model using Self-Attention Distillation (SAD) to maintain real-time performance of 55+ FPS on resource-constrained embedded hardware.

Management Apps For Inventory PT. Mitra Cerah Cemerlang

- A full-stack B2B ERP and Accounting platform designed to automate livestock feed distribution and inventory tracking.
- Built with a Flutter mobile/web interface and a scalable Firebase cloud backend.

House Price + Salary Analysis Project using Random Forest Algorithm

- A data science project utilizing Random Forest algorithm to predict real estate market based on regional economic data.
- Built a comparative predictive framework using Random Forest and XGBoost, reaching a maximum prediction accuracy of 91.35%.

SIBI Recognition Using CNN-GRU to Assist Deaf People in Greetings and Introductions

- A project designed to recognize Indonesian Sign Language (SIBI) greetings in real-time by using camera.
- Developed a spatial-temporal deep learning model integrating CNN and GRU, achieving 97.8% accuracy in recognizing dynamic sign language gestures.

PUBLICATION

Alvan, S. (2026). IndoBERT-Based Extractive Summarization with K-Means Clustering and TextRank Re-ranking for Indonesian Academic Lecture Transcripts. The 3rd International Conference on Technology Innovation and Its Application (ICTIIA 2026), Surabaya, Indonesia.

Siburian, A., Samosir, F., & Alvan, S. (2026). Latent Diffusion for Cross-Lingual Syntax Transfer in Distant Language Pairs. The 9th International Conference of Computer and Informatics Engineering (IC2IE), Depok, Indonesia.

ORGANIZATION EXPERIENCE

Mentor of Food Technology 2024-2025 — **June 2024 - August 2025**

- Mentored new students on academic adjustments and core curriculum requirements.

Algorithm & Data Structure Tutor of TutorME Even Semester 2025 — **February 2025**

- Teaching Underclassmen efficient programming logic, Big O analysis, and Algorithm in Data structures.

Steering Committee of Comparative Study UPHx BEM FASILKOM UI — **August 2024 - September 2024**

- Supervised the executive board and providing approval for event operations and planning.

Steering Committee of Weltrip 2024 — **July 2024 - October 2024**

- Supervised the executive board and providing approval for event operations and planning.

Coordinator of Logistic Falcon Project 13 — **January 2024 - April 2024**

- Managing requested rooms and equipment distribution to ensure event operations.

Calculus 1 Tutor of TutorME 2024 — **October 2023**

- Teaching Underclassmen fundamental mathematical concepts in Calculus 1.